Manish Kumar

AI Researcher & Software Engineer

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SUMMARY

Software Engineer and AI Researcher focused on generative architectures, large-scale video models, and educational AI. Specialize in bridging the gap between state-of-the-art open-source research and production-grade applications. Passionate about scaling deep learning training pipelines and building high-throughput inference architectures. Seeking a full-time engineering role to solve complex, large-scale machine learning challenges.

EXPERIENCE

Lead AI Engineer *Dec 2025 – Present*
Zulense, Faridabad, Haryana

- Key developer and architect behind an innovative, vertical video foundation model designed for pedagogical AI applications.
- **AI Architecture:** Architected the Zulense Z1 Model, focusing on text-to-video generation tailored for digital educational content (e.g., automated whiteboard handwriting and step-by-step visualization sequences).
- **Infrastructure Scaling:** Developed proprietary, distributed training infrastructure and optimized compute scaling for efficient model training utilizing Google Cloud Platform (GCP) and Google AI Studio.
- **Technical Strategy:** Aligned product requirements with cutting-edge engineering pipelines, ensuring robust system stability and low-latency API endpoints.

Volunteer Open-Source Contributor *Sep 2025 – Feb 2026*
Hugging Face (Diffusers Library), Remote

- Contributed core improvements to the industry-standard PyTorch ecosystem for generative image and video models.
- **Pipeline Optimization:** Improved inference latency and significantly reduced VRAM consumption for diffusion models, leading to smoother runtime executions.
- **Core Architecture:** Researched, tested, and implemented custom U-Net designs and Denoising Diffusion Probabilistic Models (DDPM).
- **Collaboration:** Authored and reviewed high-impact pull requests focused on code refactoring, pipeline stability, and documentation.

Open-Source AI Researcher *Jul 2024 – Sep 2025*
LAION, New Delhi, Delhi

- Conducted applied research focusing on generative architectures, representation learning, and advanced Transformer models.
- **Transformer Research:** Explored and implemented advanced Transformer architectures designed for generative AI environments.
- **Data Engineering:** Analyzed and curated large-scale multimodal datasets to understand complex text and image representations for training foundation models.

Software Engineer (Independent Consultant) *Feb 2023 – Jul 2024*
Gurugram, Haryana

- Specialized in full-stack web and cross-platform mobile development, delivering high-performance,

- scalable software solutions.
- **Mobile & Backend:** Developed responsive mobile applications using React Native and built highly secure, scalable RESTful APIs with Node.js.
- **State & UI Management:** Implemented complex state management solutions and designed responsive UI/Frontend elements for production e-commerce applications.

EDUCATION

Bachelor of Computer Applications (BCAOL) *Feb 2023 – Feb 2027 (Expected)*
IGNOU, Delhi

- *Note: Distance/Online program with fully asynchronous, flexible scheduling—entirely compatible with full-time professional employment commitments.*
- **Core Coursework:** Data and File Structures, DBMS, Object-Oriented Technology (C++ & Java), Algorithm Design & Problem Solving, Computer Architecture.

12th Grade (Science) *Feb 2019 – Feb 2021*
Bihar School Examination Board (BSEB)

- Score: 368/500 | Subjects: Physics, Chemistry, Biology, English, Hindi.

KEY PROJECTS & OPEN SOURCE

- **Hugging Face Zulense Z1 Model:** Foundational vertical video model architecture designed for automated, sequential educational visualizations.
- **Hugging Face Whiteboard Datasets:** Curated custom text-to-video datasets designed to optimize sequence-based visual learning.
- **GitHub `stable-diffusion-from-scratch`:** Clean, educational engineering breakdown of text-to-image synthesis pipelines built from the ground up.
- **GitHub `causal-vae` & `video_generation`:** Spatiotemporal modeling pipelines and core visual synthesis architectures.
- **GitHub `DDPM` & `VQ-VAE`:** End-to-end implementation of probabilistic models and discrete latent spaces.

TECHNICAL SKILLS

- **AI & ML:** Video Diffusion Models, PyTorch, Hugging Face Diffusers, U-Net, Transformers, DDPM, VQ-VAE, Causal VAE.
- **Development:** Python, Node.js, React Native, C++, Java, Full-Stack Web Development, HTML/CSS.
- **Tools & Cloud:** Google Cloud Platform (GCP), Google AI Studio, Git, GitHub Ecosystem, CI/CD.